

AdaTransolver: Measure-Aware Spatial Test-Time Scaling for Neural Operators

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Abstract

Neural operators can accept irregular point sets, but this flexibility does not guarantee that a frozen model represents the same continuous operator after the sampling density changes. Local densification changes the empirical measure used by point aggregation and can therefore make additional test-time samples harmful. We formulate spatial resolution as a test-time compute resource and introduce AdaTransolver, a framework that combines variable-point-count training, measure-aware Physics-Attention, and adaptive spatial allocation. The model is trained on uniformly sampled subsets with varying cardinality, while a self-normalized importance correction is applied only to the point-to-slice reductions at inference. A coarse prediction, a geometry prior, or a lightweight acquisition head then allocates a fixed point budget across spatial patches. Across five standard PDE benchmarks, coarse-field allocation lowers the normalized area under the error–budget curve from the uniform reference of 1.0 to 0.574–0.937. On the NACA airfoil benchmark, it reduces relative L_2 error from 0.0169/0.0133/0.0093 to 0.0100/0.0070/0.0062 at 37.5/50/75% point budgets, respectively. A soft allocation variant obtains rAUEC = 0.705 on an 8.5-million-cell DrivAerML surface-pressure task. These results show that spatial test-time scaling is useful when the refinement signal identifies reducible local error and the frozen operator remains stable under the induced measure shift.

1 Introduction

Neural operators learn maps between function spaces and can amortize expensive numerical simulation across many physical configurations (Li et al., 2021; 2023b; Wu et al., 2024). Their practical appeal is especially strong for computational fluid dynamics and solid mechanics, where high-fidelity meshes contain far more points than can be processed during every training or inference step. Recent architectures address this scale by compressing irregular meshes into latent tokens, distributing the computation, or decoupling global state estimation from pointwise decoding (Alkin et al., 2024; Luo et al., 2025; Zhou et al., 2026). These approaches make variable input sizes computationally feasible, but they leave a separate question open: given a fixed inference budget, where should the available spatial samples be spent?

Uniform subsampling is a robust default, yet it ignores the strongly localized structure of many PDE solutions. Shocks, boundary layers, contact interfaces, and stress concentrations occupy only a fraction of the domain. A natural strategy is therefore to run a coarse prediction and move the remaining point budget toward regions with large predicted variation. This operation is not a free composition of an adaptive sampler and an irregular-grid network. Standard point aggregation uses equal-weight sums or normalized averages. Densifying one region consequently gives it more empirical mass, even if the underlying physical area or volume is unchanged. A model may accept the resulting point set while no longer approximating the same continuous map.

This distinction separates three capabilities that are often conflated: irregular-grid compatibility, which concerns whether a network accepts different meshes; measure consistency,

Variable- N training $\rightarrow$ frozen measure-aware operator $\rightarrow$ coarse subset $S_0 \rightarrow$
 field/geometry/head score $\rightarrow$ patch allocation + FPS $\rightarrow$ refined subset S_1 and reconstruction

Figure 1: Overview of AdaTransolver. Uniform random subsets with variable cardinality make one checkpoint usable across point budgets. At test time, a refinement policy re-allocates the remaining spatial budget, while measure-aware point-to-slice aggregation keeps the target physical measure fixed. The model parameters remain frozen throughout both inference passes.

which concerns whether reductions approximate the same physical integral under different sampling densities; and test-time adaptive allocation, which concerns whether extra spatial compute is directed to useful regions. Prior work has made substantial progress on each component separately (Liu et al., 2023; Berner et al., 2026; Doherty et al., 2024; Gao et al., 2025; Rowbottom et al., 2024), but their combination remains underexplored.

We study this intersection through AdaTransolver. The framework first trains a single Transolver backbone on uniformly sampled subsets whose cardinality varies from iteration to iteration. At inference, it corrects every point-to-slice reduction with a relative cell-mass weight, preventing locally dense regions from being counted repeatedly. A refinement policy then performs a coarse pass, scores local field variation, aggregates the scores into approximately equal-measure patches, and redistributes the remaining budget under an explicit density-ratio constraint. The refined pass uses the same frozen network parameters. Geometry priors and a lightweight learned acquisition head instantiate the same policy interface, allowing the source and cost of the refinement signal to be studied independently from measure correction.

Our contributions are as follows.

- We formulate spatial test-time scaling for neural operators and identify sampling-measure mismatch as the obstacle that prevents irregular input support from directly enabling local refinement.
- We introduce a measure-aware Physics-Attention rule that inserts self-normalized importance weights only into point-to-slice aggregation. It reduces to the original model for uniform weights and requires no change to slice attention, deslicing, or pointwise layers.
- We combine variable-cardinality training with policy-decoupled refinement based on coarse-field gradients, geometry priors, or a learned acquisition head. Patch-level allocation, a density cap, and farthest-point placement maintain global coverage.
- We evaluate the resulting accuracy–budget frontier on five standard PDE benchmarks and an industrial automotive surface task. We also report negative cases that expose when error is unpredictable, irreducible by sampling, or too spatially diffuse for local refinement.

2 Related Work

Neural operators on irregular geometries. Fourier neural operators learn mesh-independent spectral kernels on regular grids, and Geo-FNO extends this idea to general geometries through learned deformations (Li et al., 2021; 2023a). GINO couples graph-based integral operators with a latent grid, while NUNO partitions non-uniform observations to control interpolation error (Li et al., 2023b; Liu et al., 2023). Transformer-based operators such as GNOT, UPT, GAOT, and the Transolver family compress large point sets into latent tokens before global interaction (Hao et al., 2023; Alkin et al., 2024; Wen et al., 2025; Wu et al., 2024). Transolver++ and Transolver-3 further scale this pattern to million- and hundred-million-cell geometries (Luo et al., 2025; Zhou et al., 2026). These methods establish compatibility with irregular inputs and changing resolution. Our focus is narrower but complementary: whether actively changing the local sampling density preserves the continuous operator and improves a frozen checkpoint under a finite budget.

Discretization-consistent operator learning. Extending finite neural architectures to function spaces requires their reductions to converge to continuous integrals as the discretization changes (Berner et al., 2026). Monte Carlo convolution corrects non-uniform point-cloud sampling with inverse-density weights (Hermosilla et al., 2018), while QuadConv evaluates learned continuous kernels through numerical quadrature on irregular PDE data (Doherty et al., 2024). Discretization mismatch can remain substantial even for architectures whose parameters do not depend on grid resolution (Gao et al., 2025). Related work also studies discretization-invariant maps between neural fields and quadrature-aware normalization (Wang & Golland, 2022; Kim et al., 2026). We apply the same functional principle to Physics-Attention, but use it to enable an active test-time intervention: the sampling density is intentionally changed to improve the accuracy-compute frontier.

Adaptive meshes and test-time refinement. Classical adaptive discretization reduces numerical error by refining elements whose residual or estimated contribution is large. Learned variants predict mesh density or node relocation for numerical solvers (Huang et al., 2021; Obiols-Sales et al., 2023; Rowbottom et al., 2024). MeshGraphNets similarly operates on mesh graphs and supports remeshing in learned simulation (Pfaff et al., 2021). These methods primarily use adaptivity to construct a mesh for a numerical or recurrent simulator. In neural-operator test-time scaling, PDE-Refiner adds iterative denoising steps and operator splitting adds test-time operator composition (Lippe et al., 2023; Serrano et al., 2026). Ada-Transolver instead scales the spatial sample budget of one frozen feed-forward operator. This axis is orthogonal to increasing model size, pretraining data, or the number of refinement iterations.

3 Method

3.1 Problem formulation

Let Ω be a physical domain with target measure μ , and let $\mathcal{G}_\theta$ map an input function f to a solution field u . For each test instance, a high-resolution candidate discretization $\mathcal{X}_H = \{x_i\}_{i=1}^{N_H}$ is available, but the operator may be evaluated on at most $N_f < N_H$ points. The goal is to select a subset $S \subset \{1, \dots, N_H\}$, evaluate the frozen model on $\mathcal{X}_S$, and reconstruct the full field with lower error than uniform sampling at the same final point count. The end-to-end computation includes selection, operator evaluation, and reconstruction; we separately report point-budget results and identify equal-time evaluation as an outstanding requirement.

3.2 Variable-cardinality training

A checkpoint trained only at one point count can fail before any allocation policy is considered. We therefore treat every benchmark as an unstructured point cloud during training. At iteration t , we draw

$$N_t \sim \mathcal{U}(\lceil \rho_{\min} N_H \rceil, N_H), \quad S_t \sim \text{UniformSubset}(N_H, N_t), \quad (1)$$

and compute the task loss only on S_t . Standard two-dimensional benchmarks use $\rho_{\min} = 0.3$; very large three-dimensional meshes use a smaller range such as 1–10%. The subset is shared across examples in a minibatch and, for temporal tasks, across the rollout. This training rule changes the number of observations without introducing a non-uniform spatial density. It should therefore be distinguished from density-adaptive training, which did not improve the strongest NACA checkpoint in our experiments.

3.3 Measure-aware Physics-Attention

Transolver assigns each point feature h_i to G physical slices using non-negative soft assignments a_{ig} , normalized over slices for each point. The standard slice token is

$$z_g = \frac{\sum_{i \in S} a_{ig} v(h_i)}{\sum_{i \in S} a_{ig}}. \quad (2)$$

Under uniform sampling, Eq. (2) is a self-normalized Monte Carlo estimate of an integral over μ . If points are instead sampled from a density q , the same expression integrates against $q d\mu$ and overweights locally dense regions.

Let m_i denote the physical mass represented by selected point i . On meshes with cell areas or volumes, m_i is obtained directly; otherwise we estimate it by assigning every candidate point to its nearest selected point. With $\bar{m} = |S|^{-1} \sum_{i \in S} m_i$, we define

$$c_i(\beta) = \left(\frac{m_i}{\bar{m}} \right)^\beta, \quad 0 \leq \beta \leq 1, \quad (3)$$

and replace Eq. (2) by

$$z_g^{(\beta)} = \frac{\sum_{i \in S} c_i(\beta) a_{ig} v(h_i)}{\sum_{i \in S} c_i(\beta) a_{ig}}. \quad (4)$$

Here $\beta = 0$ recovers the original point-count aggregation, $\beta = 1$ is the full measure correction, and intermediate values trade finite-sample bias against variance. We select β on a validation set, typically obtaining 0.5–0.75.

The correction is deliberately local to the two sums in Eq. (4). Slice assignment remains normalized over slices, self-attention acts only on the G tokens, deslicing remains pointwise, and feature normalization acts over channels. Consequently, setting $c_i = 1$ exactly recovers the original computation. In a replication audit, full correction reduced the deviation caused by local point replication from 2.0×10^{-2} to 4.7×10^{-7} .

Consistency. Suppose x_i are sampled independently from q and $c_i \propto d\mu/dq(x_i)$. If the denominator is nonzero and the token features are integrable, the strong law of large numbers applied to the numerator and denominator gives

$$z_g^{(1)} \xrightarrow{a.s.} \frac{\int_{\Omega} a_g(x) v(h(x)) d\mu(x)}{\int_{\Omega} a_g(x) d\mu(x)}. \quad (5)$$

Thus full correction converges to a token defined by the target physical measure rather than by the chosen sampling density. This argument does not guarantee finite-sample improvement; it only removes one source of systematic measure shift.

3.4 Adaptive spatial allocation

Our primary policy uses one coarse pass and one refined pass.

Coarse prediction and local score. We draw a uniform initial subset S_0 with $N_0 = \rho_0 N_H$ points, using $\rho_0 = 0.25$ on the standard benchmarks. From the normalized coarse prediction $\hat{u}^0$, we estimate local field variation with k nearest neighbors:

$$r_i = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} \frac{\|\hat{u}_i^0 - \hat{u}_j^0\|_2^2}{\|x_i - x_j\|_2^2 + \epsilon}. \quad (6)$$

The score uses no test labels. For automotive surfaces, a soft Laplacian score is more stable than hard selection and is discussed in Section 4.4.

Patch allocation. We partition the candidate discretization into K approximately equal-measure patches with spatial clustering. For patch P_k , let A_k be its physical mass and let R_k be the mean score among coarse points in the patch. We rank patches by estimated gain per available point,

$$\eta_k = \frac{A_k R_k}{|P_k \setminus S_0| + \epsilon}, \quad (7)$$

and greedily allocate the remaining $N_f - N_0$ points. To preserve global coverage, the final patch densities $d_k = |S_1 \cap P_k|/A_k$ satisfy

$$\frac{\max_k d_k}{\min_k d_k + \epsilon} \leq \kappa. \quad (8)$$

Within each selected patch, farthest-point sampling places new points relative to the existing subset. This separates which region receives compute from how points cover that region.

Policy variants. A geometry-prior policy uses an exponential function of distance to a known boundary and skips the coarse pass. A learned policy attaches a lightweight Transolver acquisition head to first-block features and predicts a local gradient target. The latter uses only approximately one eighth of the backbone depth and is $2\text{--}4.5\times$ cheaper than a full coarse forward in the current implementations. All variants share the same measure-aware refined pass.

3.5 Evaluation metrics

For a reconstructed field $\tilde{u}$ on the full candidate mesh, we report the mass-weighted relative error

$$E(S) = \left(\frac{\sum_{i=1}^{N_H} a_i \|\tilde{u}_i - u_i\|_2^2}{\sum_{i=1}^{N_H} a_i \|u_i\|_2^2 + \epsilon} \right)^{1/2}, \quad (9)$$

where a_i is physical cell mass when available. To summarize an accuracy–budget curve, we use

$$\text{rAUEC} = \frac{\int_{\mathcal{B}} E_{\text{method}}(b) db}{\int_{\mathcal{B}} E_{\text{uniform}}(b) db}, \quad \mathcal{B} = \{0.375, 0.50, 0.75\}. \quad (10)$$

Values below one improve on equal-final-point uniform sampling. This metric does not by itself establish equal-time efficiency because adaptive policies may perform an additional coarse pass.

4 Experiments

4.1 Setup

We use the standard Transolver PDE collection: Darcy flow, elasticity, pipe flow, NACA airfoil flow, and plasticity. The tasks cover regular grids, unstructured point clouds, localized interfaces, shocks, and contact regions. The backbone is trained with variable point counts and then frozen. The main comparison holds the final point set, reconstruction method, test examples, and random seeds fixed across acquisition policies. Uniform sampling is normalized to $\text{rAUEC} = 1$. We compare geometry-prior allocation, the coarse-field score in Eq. (6), farthest-point coverage, and the learned acquisition head where available. Hyperparameters β and κ are selected on a validation split.

For industrial scale, we evaluate surface-pressure prediction on DrivAerML. Each vehicle has approximately 8.5 million surface cells. The input contains normalized positions and surface normals, and the output is standardized pressure. Current results use 50 test vehicles and a one-million-point candidate pool for allocation. [Author input needed: Verify the final train/validation/test split, random seeds, and exact pool construction against the release scripts.]

4.2 Accuracy–budget scaling across PDE benchmarks

Table 1 reports the main point-budget results. Coarse-field allocation improves on uniform sampling on all five benchmarks, with the largest reduction on NACA airfoil ($\text{rAUEC} = 0.574$) and the smallest on Darcy flow (0.937). The ordering is physically plausible: airfoil shocks, elasticity stress concentrations, and plastic contact regions are localized, whereas the dominant Darcy error is more spatially diffuse. The learned head matches the full coarse policy on airfoil and remains competitive on pipe, elasticity, and plasticity while reducing acquisition cost. Geometry priors are useful but consistently weaker than the data-dependent coarse score in this evaluation.

4.3 NACA airfoil analysis

The airfoil task provides the clearest spatial scaling curve. At point budgets of 37.5%, 50%, and 75%, uniform sampling obtains relative errors 0.0169, 0.0133, and 0.0093, respectively. Coarse-gradient allocation with $\kappa = 16$ obtains 0.0100, 0.0070, and 0.0062. The last value

Table 1: Equal-final-point accuracy–budget results. Lower rAUEC is better; uniform sampling is 1.000. Full-grid relative L_2 reports the validation quality of the variable- N checkpoint. Dashes denote unavailable learned-head results. Values should be re-exported from the final experiment registry before submission.

Benchmark	Full-grid L_2	Geometry	Coarse field	Learned head
Airfoil	0.0063	0.788	0.574	0.575
Pipe	0.0048	0.870	0.811	0.819
Elasticity	0.0100	0.893	0.707	0.775
Plasticity	0.0030	0.871	0.769	0.783
Darcy	0.0136	0.988	0.937	–

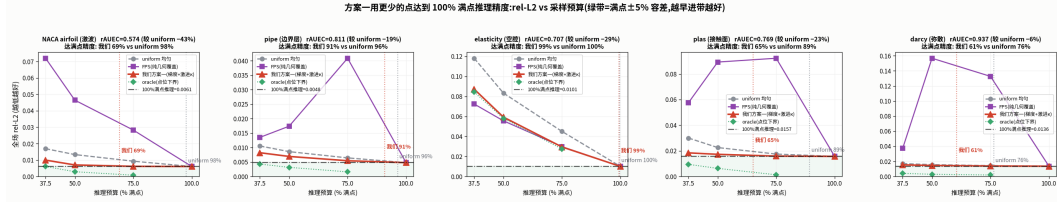


Figure 2: Error–budget curves for coarse-field adaptive allocation on five PDE benchmarks. The adaptive policy reaches the full-grid checkpoint error with approximately 69%, 91%, 99%, 65%, and 61% of the candidate points on airfoil, pipe, elasticity, plasticity, and Darcy, respectively. The embedded labels are retained from the experiment artifact and will be re-exported in English for submission.

matches the full-grid checkpoint error of 0.0061 within rounding while using one quarter fewer final points. Across five visualized test examples at a 50% budget, relative error falls by 46–69% because the refined samples concentrate on the moving shock and near-surface region (Figure 3).

The density cap controls whether the policy is sufficiently decisive. Raising κ from 4 to 16 lowers rAUEC from 0.813 to 0.574, after which performance saturates. A coverage-only farthest-point baseline is much worse on four of five benchmarks, showing that the gain does not arise merely from a more regular point set. Density-adaptive training also fails to improve the strongest NACA checkpoint. The successful combination is instead uniform variable- N training, aggressive test-time allocation, and measure correction.

4.4 Industrial automotive surfaces

Hard patch selection initially failed on DrivAerML even though an oracle reconstruction diagnostic showed that the selected locations were useful. On the same adaptive point set, oracle reconstruction error decreased from 0.153 to 0.129, but model error increased sharply. This isolates a layout shift in the frozen uniformly trained checkpoint rather than a failure of the sampling indicator. Replacing hard selection with a soft Laplacian weighting and using partial correction ($\beta = 0.5$) yields rAUEC = 0.705 for surface pressure. At 37.5%, 50%, and 75% budgets, relative error decreases by approximately 34%, 32%, and 20% against uniform sampling (Figure 4). This case illustrates why compatibility, measure consistency, and allocation should be audited separately.

4.5 Failure modes and diagnostic evidence

An auxiliary 21k-parameter error predictor clarifies when spatial scaling should not be expected to help. Learned error allocation is effective on airfoil (rAUEC = 0.763) and elasticity (0.927), approximately neutral on Navier–Stokes and plasticity, and slightly worse than uniform on pipe and Darcy (1.025 and 1.038). The failures have different causes. Navier–Stokes error is difficult to predict after chaotic rollout; plasticity has many output channels whose error regions do not coincide; pipe boundary-layer gains occupy too little of the global met-

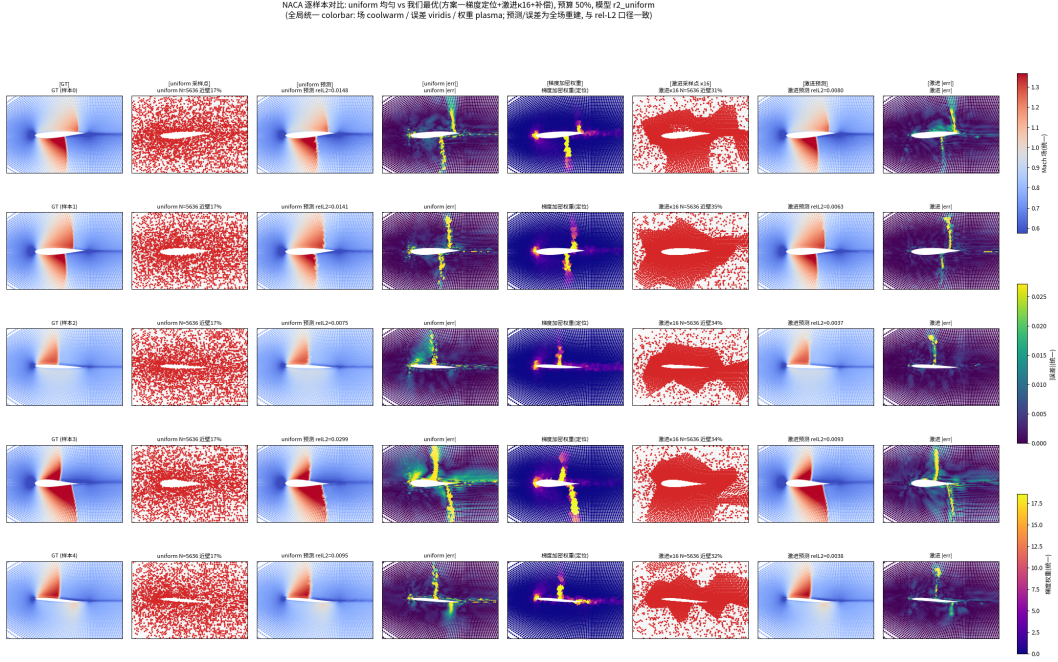


Figure 3: NACA qualitative comparison at a 50% final point budget. Columns show the reference field, uniform samples and prediction, uniform error, coarse-grained acquisition score, adaptive samples and prediction, and adaptive error. Adaptive samples track the shock location across examples. The embedded labels are retained from the experiment artifact and will be re-exported in English for submission.

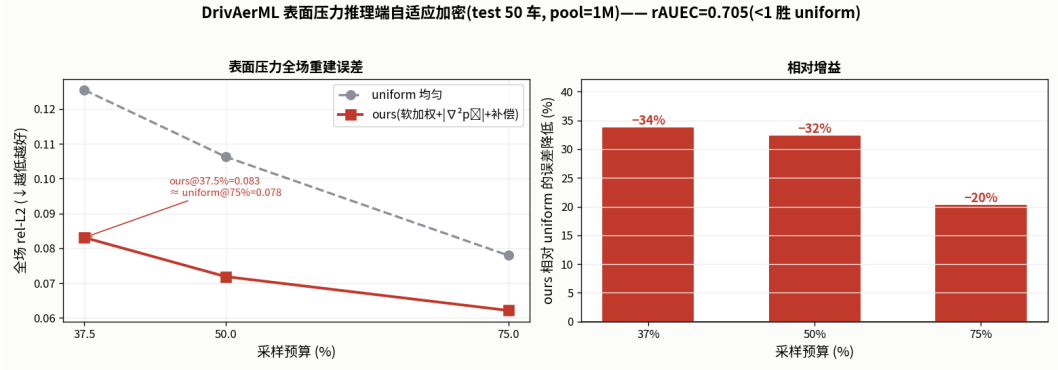


Figure 4: DrivAerML surface-pressure error under soft Laplacian allocation. The method improves all three evaluated point budgets and yields rAUEC = 0.705. The embedded labels are retained from the experiment artifact and will be re-exported in English for submission.

ric; and Darcy error is predictable but dominated by model bias that additional samples do not remove.

These observations suggest a three-factor diagnostic: adaptive sampling helps when local error is predictable from permitted test-time information, reducible by additional spatial observations, and concentrated enough to affect the global objective. Measure correction prevents a density shift from corrupting the operator, but it cannot make an uninformative acquisition score useful or remove irreducible model bias.

5 Discussion

The main result is not that one gradient heuristic solves every PDE. It is that spatial resolution can become a controllable test-time resource once the operator and the allocation policy are separated. Variable-cardinality training addresses the input-count shift, measure-aware aggregation addresses the sampling-density shift, and the policy decides where marginal resolution is valuable. This decomposition makes failures interpretable: a point-count audit, a replication audit, an oracle reconstruction audit, and a score-to-gain audit test distinct assumptions.

The evidence also exposes a finite-sample tension. Full importance correction is asymptotically aligned with the target measure, but partial correction often performs better at the point counts used in practice. We interpret $\beta < 1$ as a bias-variance trade-off, not as exact invariance. Future work should replace validation-tuned exponents with estimators that account for uncertain cell mass and token-specific effective sample size. The same principle should be audited for any spatial reduction added to the backbone, including global pooling or spatial normalization.

Our current evaluation has four important boundaries. First, the main tables equalize final point count rather than end-to-end wall-clock time; the extra coarse pass must be compared with a higher-budget one-pass uniform baseline. Second, paired confidence intervals and seed-level statistics have not yet been consolidated across all result files. Third, several figures require an English re-export and a single frozen evaluation registry. Fourth, the strongest industrial result uses a task-specific soft Laplacian policy, so broader claims about a universal hard-refinement rule would be unsupported. These limitations do not change the observed point-budget improvements, but they bound the current claim to an accuracy-sampling frontier rather than a complete latency scaling law.

6 Conclusion

We introduced AdaTransolver, a framework for spatial test-time scaling of neural operators. Its central design principle is that locally adding points should change spatial resolution without changing the physical measure represented by the model. Variable-cardinality training, measure-aware Physics-Attention, and policy-decoupled refinement together improve equal-point-budget error across five PDE benchmarks and an industrial automotive surface task. The largest gain occurs when difficult regions are localized and additional samples reduce their error; diffuse, unpredictable, or model-bias-dominated errors define the method’s present boundary. Establishing equal-time gains with consolidated uncertainty estimates is the next requirement for a full test-time compute scaling claim.

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A Draft Reproducibility Checklist

- Backbone: Transolver or Transolver-3 with unstructured input handling; variable point count sampled uniformly without replacement.
- Coarse policy defaults: $\rho_0 = 0.25$, $k = 8$ nearest neighbors, $K = 48$ patches, one refinement round.
- Point budgets: 37.5%, 50%, and 75% of the candidate discretization.
- Correction: nearest-selected-point cell mass with $\beta \in \{0, 0.5, 0.75, 1\}$ selected on validation data.
- Baselines: one-pass uniform, two-pass uniform, geometry prior, coverage-only FPS, learned acquisition head, and oracle reconstruction for diagnosis only.
- Outstanding before submission: paired bootstrap confidence intervals, end-to-end latency and memory, a final seed registry, and the official ICLR 2027 style.